

Image Analysis And Pattern Recognition For Porosity Estimation From Thin Sections

Richa^{*1}, Tapan Mukerji¹, Gary Mavko¹ and Youngseuk Keehm²

¹Stanford Rock Physics Lab, Stanford University, ²Computational and Applied Geology Lab, Seoul National University, Korea

Summary

Porosity estimation from thin sections is one of the critical steps in many rock physics and petrologic applications. The estimated porosity is a key step in determining transport properties of rocks from thin sections. Conventionally, optical point count method is used to estimate porosity of thin section. This method provides an accurate measure of porosity for thin section but it is inefficient for automated porosity estimation for a large number of thin-sections. In present study we explore different pattern recognition methods that can be used to quantify the porosity of thin-section. The porosity estimate and its uncertainty depend, amongst other things, on the image analysis and pattern recognition techniques used.

Introduction

Digital rock experiments are an upcoming research area in field of rock physics. It often involves statistical reconstruction of 3D volume from 2D thin section or SEM images followed by various numerical experiments conducted on this digital rock. Thin sections are relatively easy and cheap to acquire. One of the critical properties that affect the subsequent numerical experiments is porosity estimated from the thin sections. Reliably estimating porosity from color images is not trivial and involves various image processing and image classification techniques. The present project experiments with various pattern recognition methods for porosity estimation from images of thin section.

The thin sections are often prepared by epoxy impregnation, which gives the pore space a blue color as shown in Figure 1. The general methodology for estimating porosity of thin section requires conversion of the colored image of a thin section (Figure 1(a)) to a binary image (B&W) image as shown in Figure 1(b). In the binary image, the black color (grains) represents '0' value while white color (pores) represents '1'. Now taking an average over the binary image gives us the porosity.

For the current exercise, we worked in both RGB and HSV color-space. The RGB color model (Jain, 1989; Gonzalez and Woods, 1992) is an additive color model that defines color in terms of combination of primaries, whereas HSV color-space encapsulates information about a color in terms of tint of color (Hue), amount of grayness (Saturation) and brightness (Value). HSV color-space, in general, is

considered to be better for color identification. We would like to compare the results obtained from both the color-spaces to ascertain this fact.

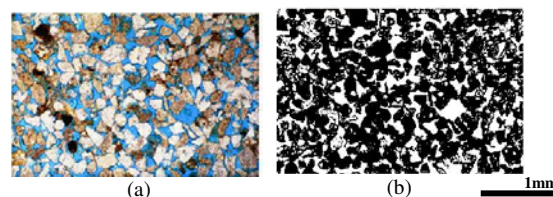


Figure 1: (a) The RGB or colored image of a thin section of a rock. The blue color represents the pore. The yellow bar denotes 1mm. (b) the binary image of the thin section in (a).

Methodology

Different approaches were tried for converting the image of thin sections to binary images. The first method used thresholds to convert to a binary image, while the second method involves classifying grains and pore space in RGB or HSV spaces using supervised learning techniques like discriminant analysis and Neural network. Lastly, unsupervised learning technique, K-mean cluster analysis were employed to classify grains and pore-space.

Thresholds:

The first method for converting the color image to a binary image uses thresholds. This can be done in two ways.

(a) Single dimension threshold: The RGB (or HSV) image can be converted to intensity image by calculating intensity value at each pixel by taking positive square root of sum of the squares of Red, Green and Blue values as given in equation (1).

$$Intensity(x) = \sqrt{R^2(x) + G^2(x) + B^2(x)} \quad (1)$$

We can calculate a single threshold value for the intensity image to convert the image to the binary image. For calculating threshold we use Otsu's method (Otsu, 1979), which chooses the threshold to minimize the intra-class variance.

(b) Multiple dimension thresholds: Three different thresholds were applied to three different dimensions of color-space (R, G and B or H, S and V). The thresholds for R, G and B were again calculated using Otsu's method. For HSV-color-space, however, Otsu's method can not be

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applied as it provides only one threshold whereas we would require more than one threshold for Hue. As the Hue describes the tint of color, the choice of thresholds was straightforward (0.5-0.75).

Supervised learning techniques:

A training data set is required to train the supervised learning technique. A small region of different category of pore and grains was selected from the thin section image. The different grains and pores within the selected regions define the RGB or HSV distributions for each type, and are collectively called the training data set (Figure 2). Two different methods, discriminant analysis and neural network were exploited. In each case, misclassification error was calculated as the overall fraction of pixels in the training data that were misclassified.

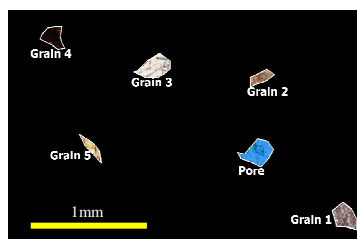


Figure 2: Training data set for thin section shown in Figure 1(a).

(a) Discriminant Analysis: Statistical discriminant analysis is used to identify boundaries between different groups of objects based on multivariate attributes of the objects (Hastie *et al.*, 2001; Krzanowski 1988; Seber, 1984). Linear, quadratic and Mahalanobis methods were used.

(b) Neural Network: A feed-forward back propagating neural network (Bishop, 1995), containing two hidden layers with 3 and 5 nodes, is used to classify the different grains and pore. It used a tan-sigmoid transfer function. Mean squared error is used to measure the network's performance.

Unsupervised learning techniques:

K-means cluster analysis (Hastie *et al.*, 2001; Seber, 1984) was used for categorizing different clusters in the image. Number of clusters, K, was varied from 7, as identified from visual inspection, to 2 for grains and pores.

Result

We present results for the thin section shown in Figure 1(a) using the different techniques discussed in previous section. The porosity for the same image obtained by visual point-count was 17.8%.

Thresholds:

(a) Single Threshold: Figure 3 shows the initial RGB image, the corresponding intensity image and the final binary version. It also shows histogram for the intensity image used for thresholding. The porosity estimated from this binary image is quite high (~50%) which implies that intra-class variance discrimination does not necessarily give the best results. Another reason for the poor result could be that when we change the RGB image to intensity image, we are averaging out the intensity of individual components (R, G, and B). This might not be the best way to approach this problem. Another problem with this mode of conversion is indicated by the red circle in the figure. The dark grain in the RGB image is taken as a pore instead of grain. As we can clearly see from the intensity image (Figure 3(b)), the pores in general have less intensity than the grains. Nevertheless, a few grains (the dark colored grains) have much lower intensity than the pores. So a single threshold value clubs these grains with the pores. Thus, different grains need to be treated separately.

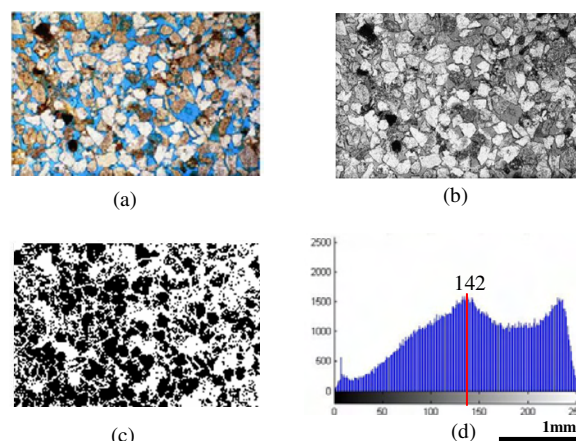


Figure 3: : Converting from RGB to binary via Intensity image using threshold: (a) RGB image; (b) intensity/gray scale image; (c) binary image; (d) histogram for the gray scale with the threshold value indicated by the red line.

For HSV color-space, the porosity obtained using this technique was more reasonable (28.8%); however the dark grain was still converted to a pore instead of a grain.

(b) Multiple thresholds: For RGB color-space, reasonable results were obtained by using thresholds for Red and Blue components only. Thresholds for Green parameter often distorted the output. The porosity obtained using this technique was 18.3%.

For HSV color-space, on the other hand, reasonable results were obtained using all the three Hue, Saturation and Value parameters. For Hue, however, two thresholds were required, in order to specify the range of blue tint. The

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typical range of hue for blue tint is 0.5-0.75. For the thresholds of Saturation and Value, Otsu's method didn't give reasonable results. The choice of the threshold for these components was based on the trial and error method. The porosity in this case was 17.9%.

The results for both RGB and HSV color-space are shown in Figure 4. The porosity obtained in each case was reasonable; however, the binary image obtained from RGB color-space was speckled as compared to HSV color-space. This becomes a problem when the quality of image is not good.

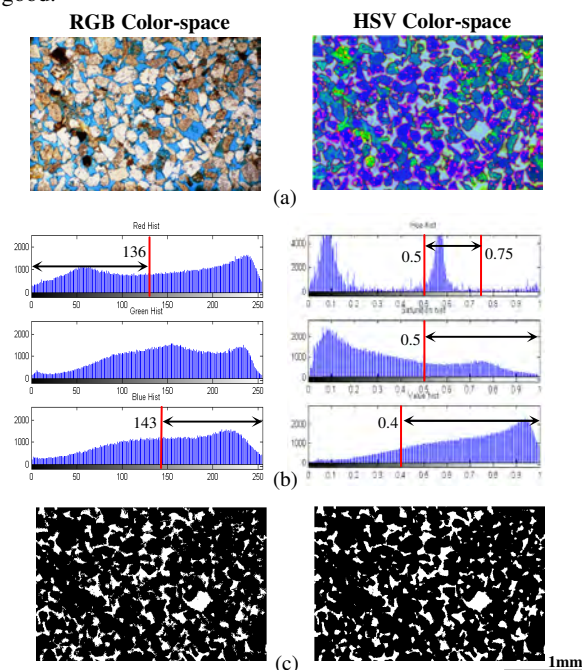


Figure 4: Multiple Dimension Threshold technique: (a) RGB and HSV Image of Thin Section; (b) Histograms of components for the image with thresholds used; (c) Binary Image obtained using Red and Blue thresholds ($\Phi=18.3\%$) and Hue, Saturation and Value thresholds ($\Phi=17.9\%$).

Supervised Learning Technique:

(a) Discriminant Analysis: The results for two different methods, linear and quadratic, are shown in Figure 5. Overall, quadratic method gave the best results and minimum error for both RGB and HSV color-space. HSV color-space again gave better results than RGB color-space.

(b) Neural Network: As observed in previous cases, HSV color-space, in general gives better results. Hence for this section we used image in HSV color-space only. The training data used to train the Neural net is same as previous section. The classification and hence the porosity obtained from different neural network or training instances

is different. The classification also depends upon the performance goal set for the neural network. The porosity variance tends to decrease with performance goal. The processing time also depends considerably on the performance goal.

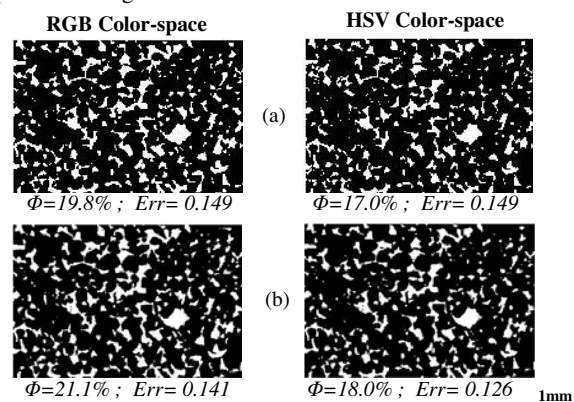


Figure 5: Discriminant Analysis: Binary images obtained by (a) Linear discriminant analysis; (b) Quadratic discriminant analysis. The respective porosities and misclassification errors are indicated below each image. The porosity obtained by point count method was 17.8%.

Figure 6 shows porosities obtained from 25 different neural networks for performance goal 0.04. The error is calculated as fraction of misclassifications if the training data is being classified using the same net. As we increase performance goal, the porosity and misclassification error as well as their variability increases. In general, errors for neural network are higher than discriminant analysis.

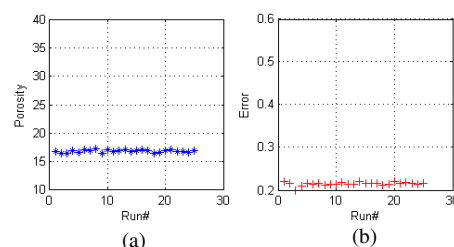


Figure 6: Neural Network: (a) Porosity and (b) error for performance goal = 0.04.

Unsupervised Learning Technique:

K-means cluster analysis was performed to test the unsupervised classification techniques. As K-means analysis, among other things, depends greatly on the number of clusters to be formed, we varied the cluster numbers from 7 to 2. Another factor that affects the result is the location of centroid for each cluster. Hence, like neural network method, the porosity estimated depends on the individual run. Two different methods were used to

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measure the distance between the individual points: squared Euclidean (L2 Norm) and City block (L1 norm). For each method, classification was carried out 25 different times. The charts for RGB and HSV color-space showing different measures of central tendency (mean, median and mode) of porosity with different cluster numbers are shown in Figure 7. In general, the porosity obtained for RGB color-space is more reasonable (<20 %) than HSV color-space.

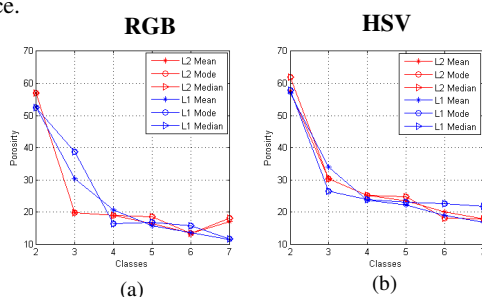


Figure 7: K-means Cluster Analysis: Variation of measures of central tendency with cluster numbers for (a) RGB Color-space (b) HSV Color-space.

Conclusions:

The HSV color-space, in general, seems to give better results than the RGB color-space for all the approaches except unsupervised classification techniques. It is obvious as HSV color-space provides a better way to identify the tint of color than RGB color-space.

The multiple threshold method works better than the single threshold method. This is evident as considering the threshold of different components separately gives us more flexibility in discriminating the exact color of the pore in the image. Using a single threshold, on the other hand gives a generalized threshold for each component. In multiple thresholds, however, the thresholds applied are subjective. Hence it provides a quick and subjective method for obtaining porosity.

The supervised learning techniques, on the other hand, provide a better way to classify the thin section image. This is particularly useful for identifying different types of grains in the thin section and hence may be useful for computational methods that require inputs beyond just porosity such as mineralogical fractions. The quadratic discriminant analysis seems to give better estimates than linear discriminant analysis. Neural network provides another way of performing statistical discriminant analysis. The results are not as good as in case of linear or quadratic discriminant. The mean and variance of the porosity and error narrows down as we decrease the performance goal. This method however requires more processing time than the linear or quadratic discriminant analysis. The problem

with supervised learning techniques is that it requires a training data set or the statistical distributions of values (RGB or HSV) for defining the pores or grains. This might require preparing a different training data for different sets of observation acquired under different camera settings.

In unsupervised learning technique, k-means cluster analysis provides subjective and run-dependent classification method. It depends greatly on number of clusters. The results are not as good as threshold methods or supervised learning techniques.

Hence, overall, multiple dimension threshold provides a fast and good, though subjective technique for estimating porosity of thin section image. The quadratic discriminant analysis is a better approach than linear discriminant analysis as well as neural network. As compared to multiple thresholds, however, it requires preparation time, although it can be useful if we require volumetric fraction of individual minerals.

Acknowledgements

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